$$= \frac{91}{6} - \frac{49}{4}$$

$$\text{Var}(X) = \frac{35}{12}$$

Example 9.11

The mean and standard deviation of a random variable X are 5 and 4 respectively Find $E(X^2)$

Solution:

Given $E(X) = 5$ and $\sigma = 4$

$\therefore$ $\text{Var}(X) = 16$

But, $\text{Var}(X) = E(X^2) - [E(X)]^2$

$16 = E(X^2) - (5)^2$

$E(X^2) = 25 + 16 = 41$

Example 9.12

A player tosses two coins, if two head appears he wins ₹ 4, if one head appears he wins ₹ 2, but if two tails appears he loses ₹ 3. Find the expected sum of money he wins?

Solution:

Let X be the random variable denoted the amount he wins.

The possible values of X are 4, 2 and –3

$P(X = 4) =$ P(getting two heads) $= \frac{1}{4}$

$P(X = 2) =$ P(getting one head) $= \frac{1}{2}$

$P(X = 3) =$ P(getting No heads) $= \frac{1}{4}$

Probability distribution is

X	4	2	-3
$P(\mathbf{X}=x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$E(X) = \Sigma x_i p_i$

$E(X) = x_1 p_1 + x_2 p_2 + x_3 p_3$

$E(X) = 4 \times \frac{1}{4} + 2 \times \frac{1}{2} - 3 \times \frac{1}{4} = 1.25$